



Sentiment and stance detection in COVID-19 vaccine discourse: a systematic review

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Abstract

Advances in machine learning have enhanced researchers' ability to detect vaccine hesitancy on social media using Natural Language Processing. Our objective in this study was to evaluate, through a systematic review of studies that employed machine learning to analyze sentiment and stance regarding COVID-19 vaccines on Twitter in assessing their methodological quality and consistency. We searched for papers published between 1 January 2020 and 31 December 2023 in PubMed, Web of Science, and Scopus. The inclusion criteria were the use of supervised machine learning to assess COVID-19 vaccine hesitancy through stance detection or sentiment analysis on Twitter/X. We categorized the studies according to a taxonomy of five dimensions: tweet sample selection approach, self-reported study type, classification typology, annotation codebook definitions, and interpretation of results. We analyzed the risk of bias in the included studies, examining whether stance detection was used to report different hesitancy trends compared to those using sentiment analysis. We identified 51 papers that were published in 36 journals. Our review found that measurement bias is widely prevalent in studies employing supervised machine learning to analyze sentiment and stance toward COVID-19 vaccines and vaccination. The reporting errors are sufficiently serious to hinder the generalizability and interpretation of these studies, making it difficult to determine whether discursive content communicates reluctance to vaccinate against SARS-CoV-2. Our findings underscore the importance of more transparent reporting of NLP methods in vaccine discourse studies. Addressing methodological shortcomings is essential to improving our understanding of vaccine hesitancy on social media.

Keywords Sentiment analysis · Stance detection · Vaccine hesitancy · COVID-19 · Supervised machine learning · Systematic review

Extended author information available on the last page of the article

1 Introduction

Since the onset of the COVID-19 pandemic, there has been considerable effort to understand public opinions towards COVID-19 vaccines and vaccination. Researchers have increasingly used advanced computational tools and approaches, such as Natural Language Processing (NLP), to study discourse on various social media platforms [1, 2]. The heightened attention to studying public health discussions on social media has been motivated by findings from analyses of sentiment and stance suggesting that there is growing polarization surrounding COVID-19 vaccination in discourse on platforms such as Twitter (officially known as X since 2023) [3], Facebook, and Instagram. These divisions reflect disagreements on the urgency, safety, and effectiveness of COVID-19 vaccination and the challenges posed by the COVID-19 “infodemic”—the overwhelming spread of inaccurate and misleading information. These disagreements, which manifest in microblogging text, have hindered public health responses and increased vaccine hesitancy, contributing to adverse population outcomes, including higher infection and mortality rates [4].

Researchers have typically employed two key NLP approaches, Sentiment Analysis and Stance Detection, to address the growing polarization in COVID-19 vaccine discourse on social media platforms. Although interrelated, these methods capture distinct yet complementary aspects of public discourse—Sentiment Analysis [5] identifies the overall emotional tone in a message (e.g., positive or negative), while Stance Detection [6] examines the text’s underlying position (e.g., favorable or unfavorable) toward vaccination. Together, these approaches provide valuable insights into the factors driving vaccine hesitancy and how attitudes toward vaccines are expressed. However, their application has revealed significant methodological challenges, including inconsistent usage and blurred distinctions between sentiment and stance, which can lead to confusion and misinterpretation [7, 8]. Precise and accurate conceptualization, categorization, and analysis of these methods are essential, as misclassification or inconsistency can reduce their effectiveness in assessing text that signals vaccine confidence.

This study aims to conduct a systematic review of research employing machine learning methods for sentiment analysis or stance detection to study the opinions toward COVID-19 vaccines and vaccination spread on Twitter. The study is limited to supervised machine learning on Twitter discourse for several reasons. First, this microblogging social media platform was a dominant source for discussing vaccines during the COVID-19 pandemic, with more than 368 million monthly users actively participating worldwide in 2022 [9]. Second, unlike other social media platforms like Instagram, Facebook and YouTube, the discourse on Twitter is limited to 280 characters. Third, supervised classifiers pose an important test of the conceptual conflation problem. This is because researchers using classifiers in a supervised machine learning context must make explicit labeling decisions during annotation, and therefore conflation is more clearly diagnosable.

By critically examining how researchers define, categorize, and codify sentiment and stance on a widely used microblogging platform that dominated vaccine-related debates during the COVID-19 pandemic, we aim to identify patterns, inconsistencies, and areas for improvement in the application of these methods. We explore whether

the objectives stated in these sentiment and stance studies align with their methodologies and evaluate how these approaches contribute to understanding and addressing vaccine hesitancy. As natural language processing tools become increasingly prevalent in medicine and public health, it is important that the methods be accurate. Misclassification due to conceptual ambiguity between sentiment and stance can distort assessments of vaccine confidence, potentially leading to misdirected interventions [4, 10, 11]. This systematic review identifies key sources of inconsistency in classification applications, offering guidance to improve the validity of supervised NLP studies in health policy and medicine, thereby contributing to an emerging body of research on best practices [6, 12, 13].

2 Data and methods

2.1 Study design

We conducted a systematic review in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [14] (protocol registered on PROSPERO ID CRD42023457939) to analyze studies investigating COVID-19 vaccine sentiment and stance on Twitter employing supervised machine learning models with Natural Language Processing (NLP). The variables included in our analysis are: (i) the year and journal of publication; (ii) the primary discipline of the publishing journal; (iii) the type of analysis self-reported by the researchers (sentiment vs. stance); (iv) the categories employed for tweet classification; (v) the definitions provided for the labels or categories used in annotation guidelines; (vi) the inclusion or exclusion criteria employed to filter irrelevant tweets and ensure relevance to the research objectives, specifically whether methods went beyond simple keyword filtering to capture contextually relevant data; (vii) whether a category representing neutrality was used and whether it was defined in the study; and, (viii) the central aspect addressed in the results (whether sentiment or stance).

2.2 Search strategy and selection criteria

In September 2024, we searched PubMed, Web of Science, and Scopus for papers published from 1 January 2020 to 31 December 2023. For each database, we searched article titles published in English using the following search terms and strategy: *((machine learning OR twitter) AND (covid-19 vaccine)) AND ((stance) OR (sentiment))*.

The studies were independently reviewed by two authors (XX1 and XX2). For a paper to be included in the review, it must have adhered to all of the following criteria: a machine learning model must have been used to classify tweets exclusively according to either sentiment or stance definitions; the conducted study must have trained the machine learning model (supervised); and the primary focus of the supervised sentiment or stance detection algorithm must have been on COVID-19 vaccines. Discrepancies or disagreements were adjudicated by a third reviewer (XX3). The PRISMA checklist is available in the Supplementary Material.

2.3 Data analysis

We systematically examined each included study to determine how sentiment and stance were conceptualized and analyzed. Specifically, we investigated whether studies using stance detection reported different vaccine hesitancy trends than those using sentiment analysis by assessing how hesitancy was measured and whether efforts were made to avoid potential sources of measurement bias. To this end, we employed the concept of vaccine hesitancy, as defined by the Strategic Advisory Group of Experts on Immunization (SAGE) of the World Health Organization (WHO). According to this definition, vaccine hesitancy refers to the delay in acceptance or refusal of vaccines, despite the availability of vaccination services [15].

Sentiment analysis typically studies a text's opinions and emotional tone toward an entity [5]. Stance analysis or detection assesses whether a text expresses a position that favors, is against, or has an alternative opinion toward a specific proposition or target [16]. Table 1 outlines the key differences between these study types.

Since sentiment and stance analyses have distinct research objectives [7], we evaluated whether sentiment analyses used labels consistent with those commonly employed in sentiment research [5] or whether clear annotation definitions aligned with sentiment-related concepts. Similarly, we assessed whether studies self-reported as stance detection employed commonly used stance detection categories or provided definitions that explicitly indicated that the annotation identified an opinion toward vaccination or vaccines [6].

We further investigated whether inconsistencies existed between the self-reported methodological approach in each study and the actual categories used to identify variations in COVID-19 vaccine hesitancy. Specifically, we examined whether studies introduced ambiguity into neutrality as a category, misaligned their findings with their stated methods, or made inferences about sentiment based on stance (or vice versa). We also collected metadata for each paper for further statistical analysis, including the title, publication year, authors, and venue. To ensure we captured all necessary data, we also inspected supplementary materials, if they were provided.

Table 1 Categories commonly employed in sentiment analyses and stance detection studies

Study type	Classification categories	What categories measure
Sentiment	The discourse was classified using categories that express feelings, such as “positive,” “negative,” and sometimes “neutral”	Sentiment categories are often described with “positive” reflecting optimistic or supportive emotions (e.g., expressing trust or relief) and “negative” indicating adverse emotions (e.g., fear, anger, sadness)
Stance detection	Discourse was classified using opinion-based categories, such as “pro-vaccine,” “anti-vaccine,” “neutral,” or “conditionally supportive”	For each stance category, opinions are classified considering that “pro-vaccine” indicates endorsement of vaccination, while “anti-vaccine” signifies opposition

Next, we evaluated the extent to which studies conflated sentiment and stance detection [7]. For example, we assessed whether researchers claiming to analyze sentiment focused on the tone and emotionality of discourse in relation to COVID-19 vaccine hesitancy. Similarly, we examined whether studies describing their analysis as stance detection concentrated on understanding positions or opinions toward COVID-19 vaccines and vaccination, without confusing stance with sentiment. However, we observed that the two were conflated in some cases. For example, studies sometimes inferred conclusions from sentiment that were not aligned with their targeted scope, such as equating negative sentiment with a reluctance to vaccinate. While negative sentiment captures emotional expressions of dissatisfaction or fear, it could also represent feelings of sadness resulting from a lack of vaccines or reduced vaccination coverage, for example. Similarly, inferring emotional tone from stance categories (e.g., classifying an unfavorable stance as inherently negative or emotional) introduces interpretive biases that can distort study conclusions. To address this, we classified such studies as exhibiting measurement bias.

We identified bias across studies by systematically analyzing classification categories, the use of neutrality, potential sources of measurement bias, and the filtering of irrelevant tweets. The following section details these findings, highlighting how variations in classification choices influenced interpretations of COVID-19 vaccine hesitancy on Twitter.

3 Results

The search in the selected databases retrieved 309 papers: 72 from PubMed, 43 from Scopus and 194 from Web of Science. After 84 duplicates were removed, 225 papers remained for screening. Titles and abstracts were reviewed in the first phase. Based on screening these two categories, 123 studies were excluded. In the second phase, we conducted a full-text review of 102 papers. Following the second-stage review, 51 articles were included in the systematic review, and 51 were excluded, mainly due to being unsupervised studies ($n=25$). Figure 1 illustrates the systematic review summary.

The 51 identified papers were published in 36 journals. The highest number of publications occurred in 2022, with 23 papers (41.5% of the sample), followed by 2023, with 18 papers (35.3%), and 10 papers in 2021 (19.6%). No related studies were published in 2020. The journals with more than one paper published are the Journal of Medical Internet Research ($n=8$), International Journal of Environmental Research and Public Health ($n=5$), Frontiers in Public Health, IEEE Access, International Journal of Advanced Computer Science and Applications and Social Network Analysis and Mining ($n=2$). Considering all the identified papers, the primary discipline of the journals in which these papers were published was Computer Science (37.2%), followed by Public Health and Medicine (31.3%) and Digital Health (27.5%). Studies published in a Multidisciplinary journal (4%) were the least frequent.

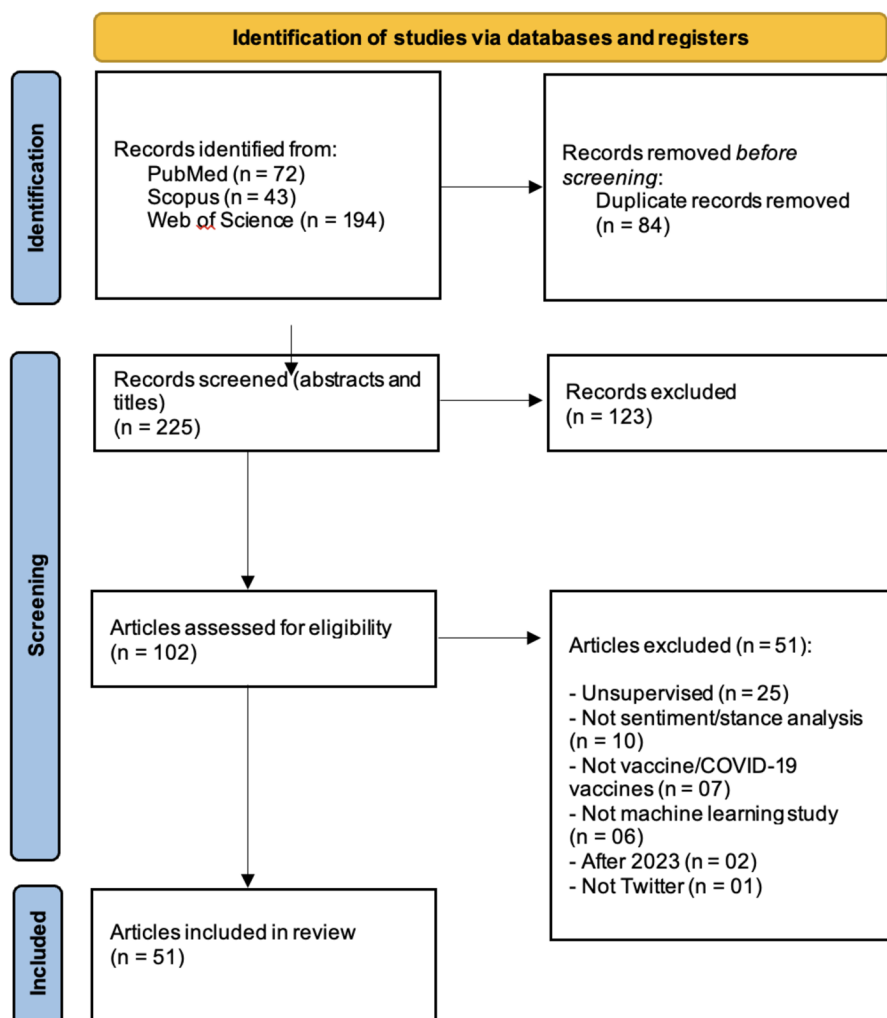


Fig. 1 Flowchart of the systematic review

3.1 Self-reported study type

Among the 51 studies, the authors described 39 (76.5%) as sentiment analysis studies, while only 12 (23.5%) were reported as stance detection studies. Studies on sentiment were most frequent in 2022 (n=21), with stance detection studies comprising only two (8.7%) publications. By contrast, stance studies represented one-third of the total publications in 2023. Regardless of the journal's disciplinary area, most studies were self-reported as sentiment analyses, with the highest share in Computer Science (94.7%). There was a relatively higher share of self-reported stance papers in Digital Health (21.4%) and Public Health and Medicine (37.5%). Table 2 details these results.

Table 2 Distribution of studies according to discipline of the journals and study type

Discipline	Journals	Sentiment (n = 39)	Stance (n = 12)
Computer science(n = 19; 37.2 ^b %)	ADCAIJ: Advances in Distributed Computing and Artificial Intelligence Journal (1) ^a	18 ^a (46.2%) ^b	1 (8.3%)
	Array (1)	(94.7%) ^c	(5.3%)
	Computation (1)		
	Computers, Environment and Urban Systems (1)		
	Concurrency and Computation: Practice and Experience (1)		
	Data in Brief (1)		
	Expert Systems with Applications (1)		
	Informatica (1)		
	International Journal of Advanced Computer Science and Applications (2)		
	International Journal of Applied Earth Observations and Geoinformation (1)		
	Journal of Intelligent and Fuzzy Systems (1)		
	Journal of Intelligent Information Systems (1)		
	Mathematics (1)		
	PeerJ Computer Science (1)		
	Social Media+ Society (1)		
	Social Network Analysis and Mining (2)		
	The Journal of Supercomputing (1)		
Digital health (n = 14; 27.5%)	American Medical Informatics Association -AMIA Annual Symposium Proceedings Archive (1)	11 (28.2%)	3 (25%)
	BMJ Health & Care Informatics (1)	(78.6%)	(21.4%)
	Digital Health (1)		
	IEEE Journal of Biomedical and Health Informatics (1)		
	JMIR Medical Informatics (1) JMIR Public Health and Surveillance (1)		
	Journal of Medical Internet Research (8)		
Multidisciplinary(n = 2; 4%)	IEEE Access (2)	0 (0%) (0)	2 (16.6%) (100%)
Public health and medicine (n = 16; 31.3%)	Bulletin of the World Health Organization (1)	10 (25.6%)	6 (50%)
	Disaster Medicine and Public Health Preparedness (1)	(62.5%)	(37.5%)
	Frontiers in Medicine (1)		
	Frontiers in Public Health (2)		
	Healthcare (1)		
	Healthcare (Basel) (1)		
	International Journal of Environmental Research and Public Health (5)		
	Journal of Behavioral Medicine (1)		
	PLOS ONE (1)		
	Vaccines (1)		

^aNumber of published papers; ^bThe percentages in parenthesis refer to the share related to the total in the column. ^cThe percentages in parenthesis refer to the share related to the total in the row

3.2 Classification typology

Table 3 confirms that the categories used to classify tweets varied, particularly in stance detection studies. Sentiment studies primarily employed “positive,” “negative,” and “neutral” categories, while stance studies included “pro-vaccine,” “anti-

Table 3 Categories employed in self-reported sentiment and stance detection studies related to COVID-19 vaccination

Type of self-described analysis	Categories employed	References
Sentiment (n = 39)	Positive, negative, neutral (n = 25)	Umair et al. [17]
		Eom et al. [18]
		Qorib et al. [1]
		Mermer et al. [19]
		Nezhad et al. [20]
		Agrawal et al. [21]
		Guo et al. [22]
		Choi et al. [23]
		Ogbuokiri et al. [24]
		Aygun et al. [25]
		Portelli et al. [10]
		Ong et al. [26]
		Kaushal et al. [27]
		Aslan [28]
		Hussain et al. [29]
		Gbashi et al. [30]
	Weakly positive, mild positive, strongly positive, weakly negative, mild negative, strongly negative, neutral (n = 1)	Arya et al. [31]
		Shahzad et al. [32]
		Alabrah et al. [33]
		Narasamma and Sreedevi [34]
		Jain and Kashyap [35]
		Vishwakarma and Chugh [36]
		Dupuy-Zini et al. [37]
		Hussain et al. [38]
		Umair et al. [39]
	Negative (without vaccine rejection causes and tweets with a rejection cause), neutral, positive (n = 1)	Alotaibi et al. [40]
		Melton et al. [41]
		Zulfiker et al. [42]
		White et al. [43]
		Qorib et al. [44]
		Roy and Ghosh [45]
		Akpatsa et al. [46]
		Jain et al. [35]
		Ljajic et al. [47]
		Chen et al. [48]
	Irrelevant, negative, positive, neutral (n = 1)	
	Negative, off-topic, positive, positive but dissatisfaction, neutral (n = 1)	
	Anti-vaccine, pro-vaccine, neutral (n = 1)	Yin et al. [49]
		Chen and Crooks [50]
	Anti-vaccine, pro-vaccine, neutral, unrelated to the topic (n = 1)	Alhumoud et al. [51]
	Acceptance, confidence, online information environment, perceived barriers to accessing vaccines (n = 1)	Zhou et al. [52]

Table 3 (continued)

Type of self-described analysis	Categories employed	References
Stance (n = 12)	Against, in favor, neutral (n = 4)	Cotfas and Delcea [53] Cotfas et al. [54] Cotfas et al. [55] Kovacs et al. [56]
	Anti-vax, pro-vax, neutral (n = 1)	Zaidi et al. [57]
	Mixed, negative, positive, neutral (n = 1)	Hwang et al. [58]
	Favorable, unfavorable, neutral (n = 1)	Jing et al. [59]
	Intent to accept COVID-19 Vaccination; intent to reject COVID-19 vaccination; belief that COVID-19 vaccines are effective; Belief that COVID-19 vaccines are unsafe (n = 1)	Zhou et al. [60]
	Promotional, neutral, discouraging, ambiguous, indiscernible (n = 1)	Cheatham et al. [61]
	Agreeing, disagreeing, no stance (n = 1)	Weinzieri et al. [62]
	Anti-vaccine, other (n = 1)	To et al. [63]
	Negative, neutral, positive (n = 1)	Lindelof et al. [64]

vaccine,” and other nuanced labels. Even though stance detection studies represent a smaller proportion of the research sample (23.5%), there is a greater variety in the category labels.

3.3 Mismatch between categories and self-described research type

The categories used to measure COVID-19 vaccine hesitancy should be mutually exclusive and non-overlapping to enhance reliability and validity. However, we identified considerable ambiguity in both sentiment and stance classification. Five (12.5%) of the sentiment studies employed stance-like categories. Two of these self-reported sentiment studies adopted the classification of tweets as anti-vaccine, neutral, or pro-vaccine, and one study [51] used the same categories and added a fourth class, denoted as unrelated. Another sentiment analysis study used the negative category to identify tweets with and without vaccine-rejection causes [40]. Finally, one self-described sentiment study adopted the categories of acceptance, confidence, and perceived barriers to accessing vaccines [52]. In contrast, two stance studies (16.7%) use sentiment analysis-related categories, such as “positive” or “negative” [58, 64]. Figure 2 highlights these findings with a Venn diagram that illustrates the overlap in categories used across sentiment and stance studies.

3.4 Missing definitions and codebook availability

Operational definitions and decision rules that guide human annotators in labeling training data into distinct categories are a core element of supervised machine learning approaches to natural language processing. The quality of these rules directly shapes model performance, as the precision and consistency of human annotations establish the benchmark against which algorithmic accuracy is evaluated. Studies

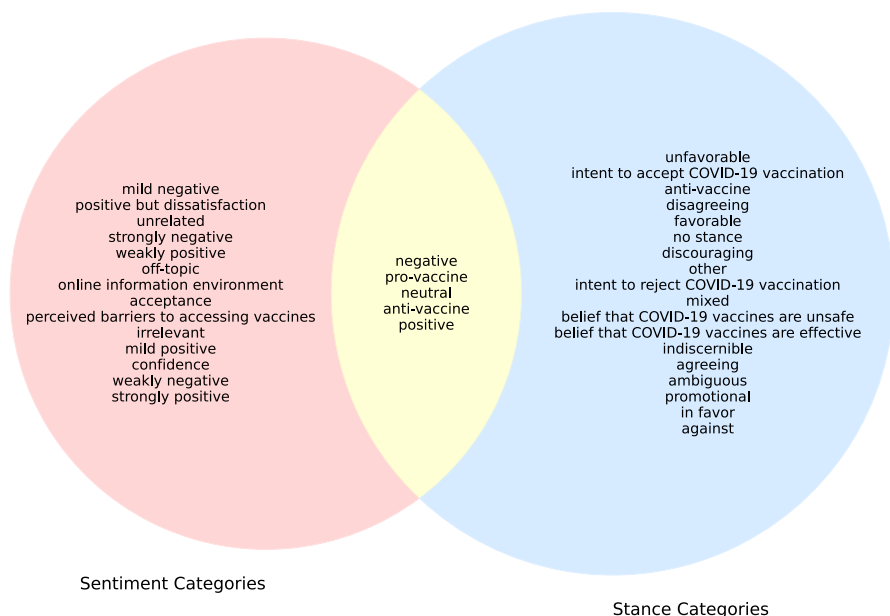


Fig. 2 Overlap (yellow) in categories used across self-reported sentiment (pink) and stance studies (blue). Categories such as ‘pro-vaccine,’ ‘anti-vaccine,’ and ‘neutral’ are common to both study types, reflecting ambiguity in classification

that lack explicit category definitions risk inconsistent and subjective labeling, introducing systematic bias into the training data and undermining the validity of supervised experiments. Only 6 (15%) sentiment analysis studies provided operational definitions. In contrast, 10 (91%) stance detection studies included explicit category definitions. Most studies ($n=35$, 68.6%) failed to provide these operational rules in the manuscript or as an appendix, such as a codebook. It is possible that authors employed definitions but did not provide them in the papers. Even if this is the case, a study with missing definitions reduces the research's reproducibility, making it harder to assess the validity of its results and, for this reason, we tabulated this attribute for each study.

3.5 Filtering of irrelevant tweets

We investigated whether sample selection bias could be identified due to systematic differences between the sampling frame and the target population in each study. To identify whether this source of bias was present, we examined whether studies reported the inclusion or exclusion criteria employed to filter out irrelevant tweets and ensure relevance to the research objectives, specifically whether methods went beyond simple keyword filtering to capture contextually relevant data. We identified only 11 (28.2%) sentiment studies and 5 (41.7%) stance studies that explicitly described their methods for removing irrelevant tweets after keyword filtering. Furthermore, many studies lacked clear documentation of how irrelevant tweets were

filtered from the selected sample, which may have introduced noise and reduced classification reliability.

3.6 Measuring neutrality and its association with vaccine hesitancy

In some studies, researchers interpreted the absence of a strong opinion in the text as an indicator of vaccine hesitancy, effectively classifying undetermined text as associated with a specific position [65]. To assess how widespread and consequential this practice was, we examined the treatment of neutrality across studies along several dimensions: whether studies provided explicit definitions of neutrality, whether these definitions linked neutral text to vaccine hesitancy, and whether neutrality was clearly distinguished from adjacent categories such as negative sentiment or an unfavorable stance.

Neutrality was included as a category in 41 (80.4%) studies. Of this total, 31 (75.6%) were sentiment-related studies [1, 10, 17–21, 23–37, 39, 40, 47, 48, 50, 51, 66]. In these sentiment analyses, neutrality was defined in only four studies, and in all four, it was defined as a lack of explicit attitude toward COVID-19 vaccination (i.e., it was defined in terms of stance) [37, 47–49]. In the other 10 (24.4%) studies, which account for 83.3% of the stance detection studies [53, 54, 56–59, 61, 63, 64, 67], neutrality was defined as news-bearing, ambiguous, or lacking strong opinions about COVID-19 vaccination [56, 58]. Finally, 29 (70.7%) studies included a neutral category but did not define it, leaving its interpretation unclear [10, 17–36, 39, 40, 44, 50, 51, 56, 58, 66].

3.7 Conflation of sentiment with stance in interpretation

Even though the objectives of sentiment and stance studies are distinct, researchers often emphasized correlations between sentiment and stance toward COVID-19 vaccines. We identified this more frequently in sentiment studies ($n = 17$, 43.5%). Thus, sentiment studies frequently conflated emotional tone with positional stance. For instance, negative sentiment was often equated with vaccine hesitancy, despite representing diverse emotional expressions, introducing interpretive bias and potential overgeneralization in conclusions. There is also one of the 12 stance detection studies (8.3%) in which the text's stance towards vaccines was asserted to be associated with sentiment. Figure 3 summarizes these findings. To a lesser extent, stance detection studies were less likely to associate positions with emotional tone. However, the sample size is smaller and this finding should therefore be interpreted with caution.

3.8 Summary of findings

To summarize our findings, Table 4 reports the frequency of the five problems that introduced bias in inferences about COVID-19 vaccine hesitancy in Twitter discourse, considering the journal's field. Adopting "neutrality" as a classification category and making claims about its association with vaccine hesitancy is a frequent practice in sentiment and stance-detection studies (80.4% of the papers). Of the 41 papers in which this phenomenon was found, 12 (29.2%) were published in public

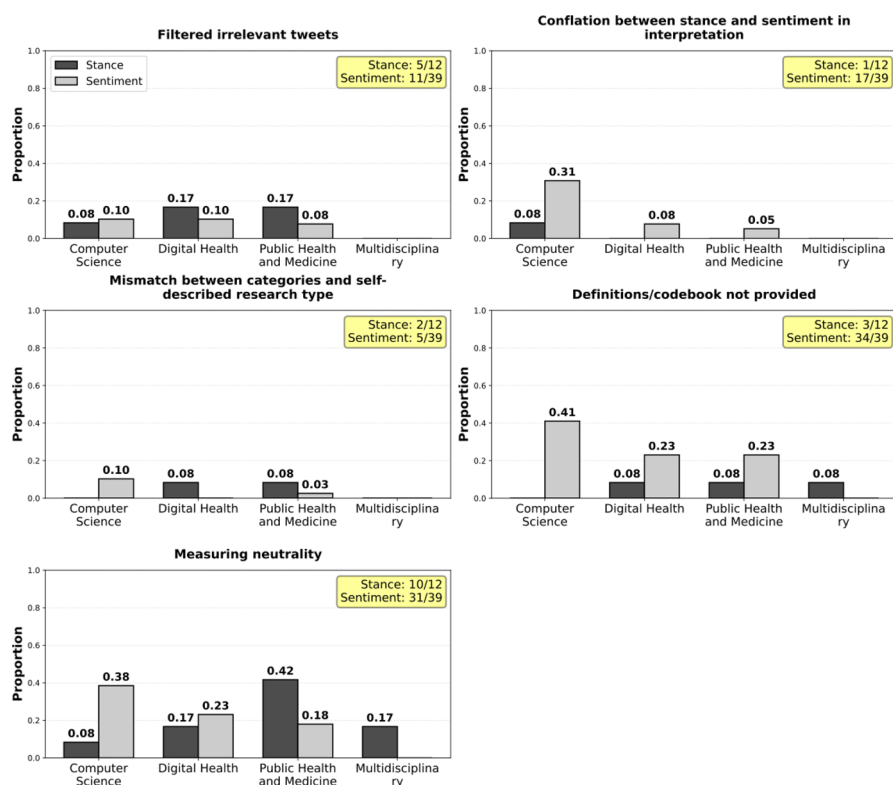


Fig. 3 Distribution of sentiment and stance studies according to filtering strategies for irrelevant tweets, conflation between stance and sentiment in interpretation, the mismatch between employed categories and self-described study type, provision of definitions/codebooks, and inclusion of neutrality as a category. (Values indicate the number of studies relative to the total examined for each classification type. The different colors in the bars represent different journal disciplines)

health, 11 (26.8%) in digital health, and 16 (39%) in computer science. However, the diverse and heterogeneous supervised machine learning techniques employed in these studies are difficult to link to the use of neutral as a classification category (see the metadata for the 51 studies reported in the Supplemental Materials).

The lack of definitions/codebooks was the second most frequent problem (72.5% of papers). Furthermore, the problem is widespread across fields: 62.5% of studies in public health, 71.4% in digital health, and 84.2% in computer science did not include detailed definitions or provide codebooks from which more detailed annotation guidelines could be retrieved.

The conflation between position and sentiment in interpretation was identified with relatively higher frequency in computer science publications (68.4%). Nevertheless, two (12.5%) studies in Public Health and Medicine and three (21.4%) studies in Digital Health also confused the position of discourse with emotional tone.

The mismatch between employing stance categories to classify emotional tone or emotional tone categories to classify stance was a less common source of bias (only

Table 4 Measurement bias by journal disciplinary area

Discipline	Filtering irrelevant tweets	Conflation between stance and sentiment in interpretation	Mismatch between categories and self-described research type	Definitions/codebook not provided	Measuring neutrality
Computer science(n= 19)	05 (31.3%) ^a (26.3%) ^b	13 (72.2%) ^a (68.4%) ^b	4 (57%) (21.0%)	16 (43.2%) (84.2%)	16 (39.0%) (84.2%)
Digital health(n= 14)	06 (37.5%) (42.9%)	3 (16.6%) (21.4%)	1 (14.3%) (7.1%)	10 (27.0%) (71.4%)	11 (26.8%) (78.6%)
Multidisciplinary(n= 2)	0 (0%) (0%)	0 (0%) (0%)	0 (0%) (0%)	1 (2.7%) (50.0%)	2 (4.9%) (100%)
Public health and medicine(n= 16)	05 (31.3%) (31.3%)	2 (11.1%) (12.5%)	2 (28.6%) (12.5%)	10 (27.0%) (62.5%)	12 (29.2%) (75%)
Overall(n= 51)	16 (31.4%) ^b	18 (35.3%) ^b	7 (13.7%) ^b	37 (72.5%) ^b	41 (80.4%) ^b

^aThe percentages in parentheses refer to the share related to the total in the column. ^bThe percentages in parentheses refer to the share related to the total in the row

13.7% of studies). Of the seven studies in which this occurred, the largest share was in computer science (57%).

4 Discussion

While opinions and sentiment towards COVID-19 vaccines and vaccination are interconnected, various disciplines have repeatedly emphasized that researchers must exercise caution when interpreting them as equivalent [7, 8]. Opinions toward vaccines are distinct from sentiment toward them, as evidenced by examples such as, “*I am so sad COVID-19 vaccines are not available for my age group yet.*” This statement expresses a negative sentiment but conveys a favorable stance. Despite this distinction, we have identified several practices in supervised machine learning studies analyzed in this systematic review that conflated sentiment with a position on vaccine hesitancy. This conflation introduces measurement bias and risks overgeneralizing conclusions, such as that negative sentiment directly corresponds to vaccine hesitancy. Although less frequent, we also identified studies conflating positions with emotional tone, such as those implying that unfavorable stances correlate with negative emotions. Recent NLP research has demonstrated that sentiment and stance often operate independently, particularly in contexts like political and social media discourse [7, 8].

This bias can be particularly problematic in public health contexts, where effective policies to address vaccine hesitancy depend on a nuanced understanding of its causes, including its detection in discourse. Vaccine hesitancy is not a dichotomous phenomenon but exists on a continuum, as explained by the World Health Organization’s (WHO) Strategic Advisory Group of Experts (SAGE) [15]. This includes individuals who delay or selectively accept vaccines at specific times, despite avail-

ability. SAGE further defines vaccine hesitancy as delaying acceptance or refusal of vaccines despite their availability. Vaccine hesitancy attitudes are influenced by various factors that may affect populations along this continuum differently [68].

Additionally, SAGE outlined vaccine hesitancy factors as the 3C's: Confidence, which involves trust in vaccine safety, health systems, and policymakers' motives; Convenience, which refers to factors like availability, affordability, accessibility, and service quality, influencing vaccine uptake; and Complacency, which arises when vaccines are seen as unnecessary due to low perceived disease risks [15]. Considering the 3C approach, sentiment and stance analysis studies on vaccine hesitancy can support vaccine promotion policies by identifying whether unfavorable positions toward vaccination are related to one of these factors. For instance, a lack of trust in pharmaceutical companies (Confidence) or the perception that vaccines are unnecessary for children due to their low risk of severe COVID-19 (Complacency) are examples of how stance analyses can uncover underlying drivers of hesitancy. This occurred, for example, in some populations regarding the perceived low risk of severe COVID-19 in children. Similarly, sentiment analysis can shed light on emotional responses related to hesitancy factors, such as frustration or sadness stemming from limited vaccine availability (Convenience), as observed during the early phases of the COVID-19 vaccination rollout. This was observed during COVID-19 vaccination when limited doses were available for all interested individuals. These emotional responses, whether negative (fear, frustration) or positive (hope, joy), provide valuable insights into public sentiment, which can be leveraged to address specific hesitancy triggers [49]. Previous studies, such as Zhao et al. [69], emphasize the role of risk perception in driving emotional responses like fear and anxiety, which subsequently influence coping behaviors during public health crises.

While analyzing sentiment and stance categories, we observed that the "neutral" category was frequently employed but often lacked clear definitions or conceptual consistency. Neutrality was often described as the absence of opinion, but this interpretation is questionable in social media discourse. Texts labeled as 'neutral' may not reflect a genuine absence of opinion (or sentiment), but rather expressions that coders found difficult to classify—ambivalent, mixed, or insufficiently explicit to assign to a favorable (or positive) or unfavorable (or negative) category. This lack of conceptual clarity has important implications for vaccine hesitancy research that relies on text classification. In most studies we reviewed, it was not clear whether neutrality was a meaningful attitudinal category or a coding residual. However, the classification of neutral discourse was sometimes linked to a study's substantive findings—for instance, by interpreting neutral expression as susceptibility to misinformation or as latent hesitancy.

As the discussion in this section has emphasized, our findings have implications for medicine and public health: NLP models that conflate sentiment and stance, or that apply poorly defined classification schemes, may introduce measurement bias into vaccine hesitancy surveillance systems that rely on text- or discourse-based detection, potentially impacting public health messaging and risk assessment efforts. It should be noted that we have outlined the implications of these problems in general terms. Nevertheless, obtaining more detailed figures would require access to the tweet samples and their corresponding annotations. Doing so would allow us to esti-

mate the magnitude of the bias as compared to the published studies reviewed in this paper. This undertaking, if at all possible, is outside the scope of this review, although it is an important question we leave for future research.

4.1 Recommendations and best practices

Considering the biases identified in sentiment and stance analysis studies of tweets about COVID-19 vaccination, several recommendations can be outlined to promote research that has greater precision in identifying vaccine hesitancy and reducing measurement error:

- (i) A clear exposition of the study's objectives—whether the goal is to identify and analyze sentiments, feelings, opinions, stances, or attitudes toward vaccines;
- (ii) The availability of comprehensive codebooks that include definitions of the categories used in the study and ensure their conceptual alignment with the study's objectives; and,
- (iii) Explicitly clarifying the conceptual and practical boundaries of the “neutral” category to avoid misclassification and ensure consistent interpretations.

In sum, the results of the systematic review revealed a surge in the number of studies dedicated to sentiment and opinion analysis of discourse on COVID-19 vaccination. However, we identified significant biases in these studies, which suggests that their reliability and applicability in shaping the understanding of how best to combat vaccine hesitancy in discourse and text analysis is limited. By addressing these issues, sentiment and stance analysis can better support public health efforts, providing actionable insights for strategies to reduce vaccine hesitancy and promote vaccination uptake.

However, this systematic review also has limitations. First, we selected sentiment and stance detection studies that utilized supervised machine learning to analyze Twitter discourse on COVID-19 vaccines and explicitly defined this objective in the title by citing one or more relevant keywords. This resulted in fewer studies than would have been identified by searching for the keywords in the abstract or text. However, the benefit of this strategy is that we ensured the study focused on research with a primary emphasis on sentiment and stance detection related to COVID-19.

Second, dictionaries are also employed in stance and sentiment classification research. In this systematic review, we excluded these studies because the conflation may be embedded in the tool itself and not clearly identifiable to the specific definitions adopted by the authors. Our study shows that bias occurs when researchers operationalize stance and sentiment constructs using explicit annotation schemes. Future research should seek to determine whether the two concepts are also conflated in studies that use dictionaries.

Third, this study focuses on Twitter, a platform that was extremely relevant in attracting users interested in microblogging globally during the COVID-19 pandemic. Future systematic reviews should assess whether our findings also apply to sentiment

and stance detection in content produced on other microblogging social media platforms. Since the majority of users on these alternative platforms are mainly concentrated in specific countries (e.g., Reddit has 50% of its users in the U.S., Weibo has primarily Chinese users), there may be significant differences that could be explored by comparing whether our research findings related to Twitter are validated.

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Data availability All data extracted and analyzed in this systematic review were derived from previously published studies. The complete dataset generated during the data extraction and analysis of the 51 included studies is available in the Supplementary Materials accompanying this article. No additional primary data were collected for this work.

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References

1. Qorib, M., Oladunni, T., Denis, M., Ososanya, E., & Cotae, P. (2023). Covid-19 vaccine hesitancy: Text mining, sentiment analysis and machine learning on COVID-19 vaccination Twitter dataset. *Expert Systems with Applications*, 212, 118715.
2. Bakhshaei, K., Rezaei, Z., Ahmadi, M., & Banad, Y. M. (2025). Pandemic transition: A review of social media text mining for pandemic transition in the post-vaccination era. *Artificial Intelligence in Medicine*, 169, 103242.
3. Allington, D., McAndrew, S., Moxham-Hall, V. L., & Duffy, B. (2021). Media usage predicts intention to be vaccinated against SARS-CoV-2 in the US and the UK. *Vaccine*, 39(18), 2595–2603.
4. World Health Organization. (2021) “WHO public health research agenda for managing infodemics”.
5. Liu, B. (2010). Sentiment Analysis and Subjectivity. In N. Indurkha & F. J. Damerau (Eds.), *Handbook of Natural Language Processing* (pp. 627–666). Chapman and Hall/CRC.
6. ALDayel, A., & Magdy, W. (2021). Stance detection on social media: State of the art and trends. *Information Processing & Management*, 58(4), 102597.
7. Bestvater, S. E., & Monroe, B. L. (2023). Sentiment is not stance: Target-aware opinion classification for political text analysis. *Political Analysis*, 31(2), 235–256.
8. Aldayel, A., & Magdy, W. (2019). *Assessing sentiment of the expressed stance on social media* Lecture Notes in Computer Science (pp. 277–286). Springer International Publishing.

9. Dixon, S. J. (2023). "X/Twitter: number of worldwide users 2019–2024," *Statista*. [Online]. Available: <https://www.statista.com/statistics/303681/twitter-users-worldwide/#statisticContainer> [Accessed: 06-Spring-2025]
10. Portelli, B., Scaboro, S., Tonino, R., Chersoni, E., Santus, E., & Serra, G. (2022). Monitoring user opinions and side effects on COVID-19 vaccines in the Twittersphere: Infodemiology study of tweets. *Journal of Medical Internet Research*, 24(5), e35115.
11. Calleja, N., AbdAllah, A., Abad, N., Ahmed, N., Albarracin, D., Altieri, E., Anoko, J. N., Arcos, R., Azlan, A. A., Bayer, J., Bechmann, A., Bezbaruah, S., Briand, S. C., Brooks, I., Bucci, L. M., Burzo, S., Czerniak, C., De Domenico, M., Dunn, A. G., ... Purnat, T. D. (2021). A public health research agenda for managing infodemics: Methods and results of the first WHO infodemiology conference. *JMIR infodemiology*, 1(1), e30979.
12. Islam, S. M. S., Tabassum, R., Liu, Y., Chen, S., Redfern, J., Kim, S. Y., Ball, K., Maddison, R., & Chow, C. K. (2019). The role of social media in preventing and managing non-communicable diseases in low-and-middle income countries: Hope or hype? *Health Policy Technology*, 8(1), 96–101.
13. Straw, I. (2021). Ethical implications of emotion mining in medicine. *Health Policy Technol.*, 10(1), 191–195.
14. "The PRISMA 2020 statement: An updated guideline for reporting systematic reviews" (2020). [Online]. Available: <https://www.equator-network.org/reporting-guidelines/prisma/> [Accessed: 18-Jul-2022]
15. MacDonald, N. E., SAGE Working Group on Vaccine Hesitancy. (2015). Vaccine hesitancy: Definition, scope and determinants. *Vaccine*, 33(34), 4161–4164.
16. Alturayef, N., Luqman, H., & Ahmed, M. (2023). A systematic review of machine learning techniques for stance detection and its applications. *Neural Computing and Applications*, 35(7), 5113–5144.
17. Umair, A., & Masciari, E. (2023). Sentimental and spatial analysis of COVID-19 vaccines tweets. *Journal of Intelligent Information System*, 60(1), 1–21.
18. Eom, G., Yun, S., & Byeon, H. (2022). Predicting the sentiment of South Korean Twitter users toward vaccination after the emergence of COVID-19 Omicron variant using deep learning-based natural language processing. *Frontiers in Medicine*, 9, 948917.
19. Mermer, G., & Özsezer, G. (2022). Discussions about COVID-19 vaccination on Twitter in Turkey: Sentiment analysis. *Disaster Medicine and Public Health Preparedness*, 17, e266.
20. Nezhad, Z. B., & Deihimi, M. A. (2022). Analyzing Iranian opinions toward COVID-19 vaccination. *IJID Regions*, 3, 204–210.
21. Agrawal, S., Jain, S. K., Sharma, S., & Khatri, A. (2022). COVID-19 public opinion: A Twitter healthcare data processing using machine learning methodologies. *International Journal of Environmental Research and Public Health*, 20(1), 432.
22. Guo, Y., Zhu, J., Huang, Y., He, L., He, C., Li, C., & Zheng, K. (2022). Public opinions toward COVID-19 vaccine mandates: a machine learning-based analysis of US tweets. In: AMIA Annual Symposium Proceedings (Vol. 2022, p. 502).
23. Choi, Y.-J., Lee, J., & Paek, S. Y. (2022). Public awareness and sentiment toward COVID-19 vaccination in South Korea: Findings from big data analytics. *International Journal of Environmental Research and Public Health*, 19(16), 9914.
24. Ogbuokiri, B., Ahmadi, A., Bragazzi, N. L., Movahedi Nia, Z., Mellado, B., Wu, J., Orbinski, J., Asgary, A., & Kong, J. (2022). Public sentiments towards COVID–19 vaccines in South African cities: An analysis of Twitter posts. *Frontiers in Public Health*, 10, 987376.
25. Aygun, I., Kaya, B., & Kaya, M. (2022). Aspect based twitter sentiment analysis on vaccination and vaccine types in COVID-19 pandemic with deep learning. *IEEE Journal of Biomedical and Health Informatics*, 26(5), 2360–2369.
26. Ong, S.-Q., Pauzi, M. B. M., & Gan, K. H. (2022). Text mining and determinants of sentiments towards the COVID-19 vaccine booster of twitter users in Malaysia. *Healthcare (Basel)*, 10(6), 994.
27. Kaushal, A., Mandal, A., Khanna, D., & Acharjee, A. (2023). Analysis of the opinions of individuals on the COVID-19 vaccination on social media. *Digit. Health*, 9, 20552076231186250. <https://doi.org/10.1177/20552076231186246>
28. Aslan, S. (2022). A novel TCNN-Bi-LSTM deep learning model for predicting sentiments of tweets about COVID-19 vaccines. *Concurr. Comput.*, 34(28), e7387.
29. Hussain, Z., et al. (2022). Artificial intelligence-enabled social media analysis for pharmacovigilance of COVID-19 vaccinations in the United Kingdom: Observational study. *JMIR Public Health and Surveillance*, 8(5), e32543.

30. Gbashi, S., Adebo, O. A., Doorsamy, W., & Njobeh, P. B. (2021). Systematic delineation of media polarity on COVID-19 vaccines in Africa: Computational linguistic modeling study. *JMIR Medical Informatics*, 9(3), e22916.
31. Arya, V., Mishra, A. K., & González-Briones, A. (2022). Sentiments Analysis of Covid-19 Vaccine Tweets Using Machine Learning and Vader Lexicon Method. *ADCAIJ: Advances in Distributed Computing and Artificial Intelligence Journal*, 11(4), 507–518.
32. Shahzad, A., Zafar, B., Ali, N., Jamil, U., Alghadhban, A. J., Assam, M., Ghamry, N. A., & Eldin, E. T. (2022). COVID-19 vaccines related user's response categorization using machine learning techniques. *Computation (Basel)*, 10(8), 141.
33. Alabrah, A., Alawadh, H. M., Okon, O. D., Meraj, T., & Rauf, H. T. (2022). Gulf countries' citizens' acceptance of COVID-19 vaccines—A machine learning approach. *Mathematics*, 10(3), 467.
34. Narasamma, V.L., & Sreedevi, M. (2021) Twitter based data analysis in natural language processing using a novel catboost recurrent neural framework. *International Journal of Advanced Computer Science and Application*. 12(5).
35. Jain, T., Verma, V. K., Sharma, A. K., Saini, B., Purohit, N., Mahdin, H., Ahmad, M., Darman, R., Su-Cheng, H., Shaharudin, S. M., & Arshad, M. S. (2023). Sentiment analysis on COVID-19 vaccine tweets using machine learning and deep learning algorithms. *International Journal Advanced Computer Science and Applications*, 14, 5.
36. Vishwakarma, A., & Chugh, M. (2023). COVID-19 vaccination perception and outcome: Society sentiment analysis on twitter data in India. *Social Network Analysis and Mining*, 13(1), 84.
37. Dupuy-Zini, A., Audeh, B., Gérardin, C., Duclos, C., Gagneux-Brunon, A., & Bousquet, C. (2023). Users' Reactions Announced Vaccines Against CO VID-19 Before Marketing France: Analysis Twitter Posts. *Journal Medical Internet Research*, 25, e37237.
38. Hussain, A., Tahir, A., Hussain, Z., Sheikh, Z., Gogate, M., Dashtipour, K., Ali, A., & Sheikh, A. (2021). Artificial intelligence-enabled analysis of public attitudes on Facebook and twitter toward COVID-19 vaccines in the United Kingdom and the United States: Observational study. *Journal of medical Internet research*, 23(4), e26627.
39. Umair, A., Masciari, E., & Ullah, M. H. (2023). Vaccine sentiment analysis using BERT + NBSVM and geo-spatial approaches. *The Journal of Supercomputing*, 79(15), 1–31.
40. Alotaibi, W., Alomary, F., & Mokni, R. (2023). COVID-19 vaccine rejection causes based on Twitter people's opinions analysis using deep learning. *Social Network Analysis and Mining*, 13(1), 62.
41. Melton, C. A., White, B. M., Davis, R. L., Bednarczyk, R. A., & Shaban-Nejad, A. (2022). Fine-tuned sentiment analysis of COVID-19 vaccine-related social media data: Comparative study. *Journal of Medical Internet Research*, 24(10), e40408.
42. Zulfiker, M. S., Kabir, N., Biswas, A. A., Zulfiker, S., & Uddin, M. S. (2022). Analyzing the public sentiment on COVID-19 vaccination in social media: Bangladesh context. *Array*, 15, 100204.
43. White, B. M., Melton, C., Zarcie, P., Davis, R. L., Bednarczyk, R. A., & Shaban-Nejad, A. (2023). Exploring celebrity influence on public attitude towards the COVID-19 pandemic: Social media shared sentiment analysis. *BMJ Health Care Inform.*, 30(1), e100665.
44. Qorib, M., Oladunni, T., Denis, M., Ososanya, E., & Cota, P. (2023). COVID-19 vaccine hesitancy: A global public health and risk modelling framework using an environmental deep neural network, sentiment classification with text mining and emotional reactions from COVID-19 vaccination tweets. *International Journal of Environmental Research and Public Health*, 20(10), 5803.
45. Roy, S., & Ghosh, P. (2021). A comparative study on distancing, mask and vaccine adoption rates from global Twitter trends. *Healthcare (Basel)*, 9(5), 488.
46. Akpatsa, S. K., Lei, H., Li, X., & Obeng, V. H. K. S. (2022). Evaluating public sentiment of Covid-19 vaccine tweets using machine learning techniques. *Informatica*, 46(1), 69–75.
47. Ljajić, A., Prodanović, N., Medvečki, D., Bašaragin, B., & Mitrović, J. (2022). Uncovering the reasons behind COVID-19 vaccine hesitancy in Serbia: Sentiment-based topic modeling. *Journal of Medical Internet Research*, 24(11), e42261.
48. Chen, N., Chen, X., & Pang, J. (2022). A multilingual dataset of COVID-19 vaccination attitudes on Twitter. *Data in Brief*, 44, 108503.
49. Yin, F., Crooks, A., & Yin, L. (2022). Information propagation on cyber, relational and physical spaces about covid-19 vaccine: Using social media and splatial framework. *Computers, Environment and Urban Systems*, 98, 101887.
50. Chen, Q., & Crooks, A. (2022). Analyzing the vaccination debate in social media data Pre- and Post-COVID-19 pandemic. *International Journal of Applied Earth Observation and Geoinformation*, 110, 102783.

51. Alhumoud, S., Al Wazrah, A., Alhussain, L., Alrushud, L., Aldosari, A., Altammami, R. N., Almurksh, N., Alharbi, H., & Alshahrani, W. (2023). ASAVACT: Arabic sentiment analysis for vaccine-related COVID-19 tweets using deep learning. *PeerJ Computer Science*, 9, e1507.
52. Zhou, X., Zhang, X., Larson, H. J., de Figueiredo, A., Jit, M., Fodeh, S., Vermund, S. H., Zang, S., Lin, L., & Hou, Z. (2024). Spatiotemporal trends in COVID-19 vaccine sentiments on a social media platform and correlations with reported vaccine coverage. *Bulletin of the World Health Organization*, 102(1), 32–45.
53. Cotfas, L.-A., Delcea, C., & Gherai, R. (2021). COVID-19 vaccine hesitancy in the month following the start of the vaccination process. *International Journal of Environmental Research and Public Health*, 18(19), 10438.
54. Cotfas, L. A., Crăciun, L., Delcea, C., Florescu, M. S., Kovacs, E. R., Molănescu, A. G., & Orzan, M. (2023). Unveiling vaccine hesitancy on twitter: Analyzing trends and reasons during the emergence of COVID-19 Delta and Omicron variants. *Vaccines*, 11(8), 1381.
55. Cotfas, L.-A., Delcea, C., Gherai, R., & Roxin, I. (2021). Unmasking people's opinions behind mask-wearing during COVID-19 pandemic—A twitter stance analysis. *Symmetry (Basel)*, 13(11), 1995.
56. Kovacs, E.-R., Cotfas, L.-A., Delcea, C., & Florescu, M.-S. (2023). 1000 days of COVID-19: A gender-based long-term investigation into attitudes with regards to vaccination. *IEEE Access*, 11, 25351–25371.
57. Zaidi, Z., Ye, M., Samon, F., Jama, A., Gopalakrishnan, B., Gu, C., Karunasekera, S., Evans, J., & Kashima, Y. (2023). Topics in antivax and provax discourse: Yearlong synoptic study of COVID-19 vaccine tweets. *Journal of Medical Internet Research*, 25(1), e45069.
58. Hwang, J., Su, M.-H., Jiang, X., Lian, R., Tveleneva, A., & Shah, D. (2022). Vaccine discourse during the onset of the COVID-19 pandemic: Topical structure and source patterns informing efforts to combat vaccine hesitancy. *PLoS ONE*, 17(7), e0271394.
59. Jiang, X., Su, M. H., Hwang, J., Lian, R., Brauer, M., Kim, S., & Shah, D. (2021). Polarization Over Vaccination: Ideological Differences in Twitter Expression About COVID-19 Vaccine Favorability and Specific Hesitancy Concerns. *Social Media+Society*, 7(3), 205630512110484.
60. Zhou, X., Song, S., Zhang, Y., & Hou, Z. (2023). Deep learning analysis of COVID-19 vaccine hesitancy and confidence expressed on Twitter in 6 high-income countries: Longitudinal observational study. *Journal of Medical Internet Research*, 25, e49753.
61. Cheatham, S., Kummervold, P. E., Parisi, L., Lanfranchi, B., Croci, I., Comunello, F., Rota, M. C., Filia, A., Tozzi, A. E., Rizzo, C., & Gesualdo, F. (2022). Understanding the vaccine stance of Italian tweets and addressing language changes through the COVID-19 pandemic: Development and validation of a machine learning model. *Frontiers in Public Health*, 10, 948880.
62. Weinzierl, M. A., Hopfer, S., & Harabagiu, S. M. (2023). Scaling up the discovery of hesitancy profiles by identifying the framing of beliefs towards vaccine confidence in Twitter discourse. *Journal of Behavioral Medicine*, 46(1–2), 253–275.
63. To, Q. G., To, K. G., Huynh, V. A. N., Nguyen, N. T., Ngo, D. T., Alley, S. J., Tran, A. N., Tran, A. N., Pham, N. T., Bui, T. X., & Vandelanotte, C. (2021). Applying machine learning to identify anti-vaccination tweets during the COVID-19 pandemic. *International Journal of Environmental Research and Public Health*, 18(8), 4069. <https://doi.org/10.3390/ijerph18084069>
64. Lindelöf, G., Aledavood, T., & Keller, B. (2023). Dynamics of the negative discourse toward COVID-19 vaccines: Topic modeling study and an annotated data set of Twitter posts. *Journal of Medical Internet Research*, 25(1), e41319.
65. Fulawka, K., Hertwig, R., & Pachur, T. (2024). COVID-19 vaccine refusal is driven by deliberate ignorance and cognitive distortions. *NPJ Vaccines*, 9(1), 167.
66. Hussain-Alkhatieb, L., Rivera Ramirez, T., Kroeger, A., Gozzer, E., & Runge-Ranzinger, S. (2021). Early warning systems (EWSs) for chikungunya, dengue, malaria, yellow fever, and Zika outbreaks: What is the evidence? A scoping review. *PLoS Neglected Tropical Diseases*, 15(9), e0009686.
67. Cotfas, L.-A., Delcea, C., Roxin, I., Ioanăș, C., Gherai, D., & Tajariol, F. (2021). The longest month: Analyzing COVID-19 vaccination opinions dynamics from tweets in the month following the first vaccine announcement. *IEEE Access*, 9, 33203–33223.
68. Kumar, D., Chandra, R., Mathur, M., Samdariya, S., & Kapoor, N. (2016). Vaccine hesitancy: Understanding better to address better. *Israel Journal of Health Policy Research*, 5(1), 2.
69. Zhao, Y., Jiang, Y., Zhang, W., & Zhu, Y. (2023). Relationship between risk perception, emotion, and coping behavior during public health emergencies: A systematic review and meta-analysis. *Systems*, 11(4), 181.

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